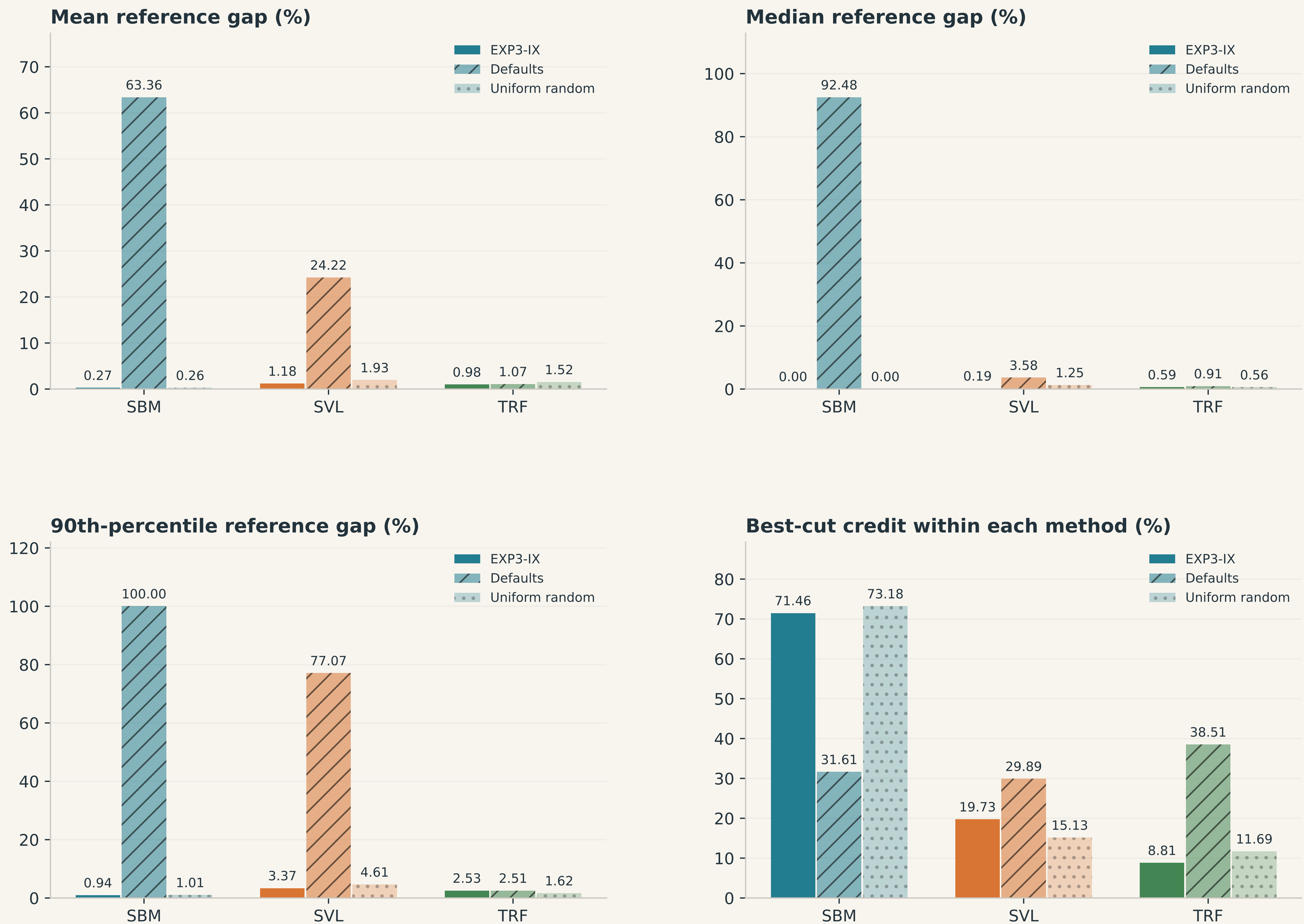
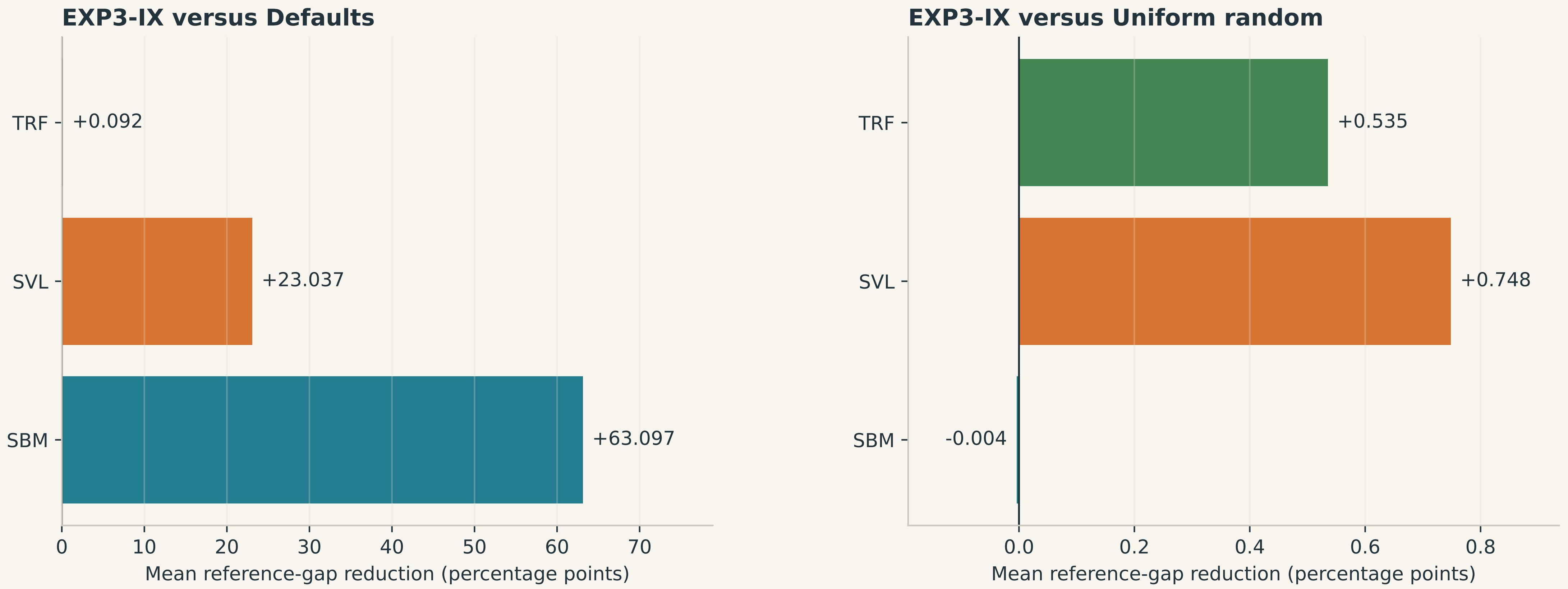
Held-out test: quality and wins

29 unseen graphs x 3 evaluation seeds. Quality ties split one point; lower gaps are better.



Did learning beat the controls?

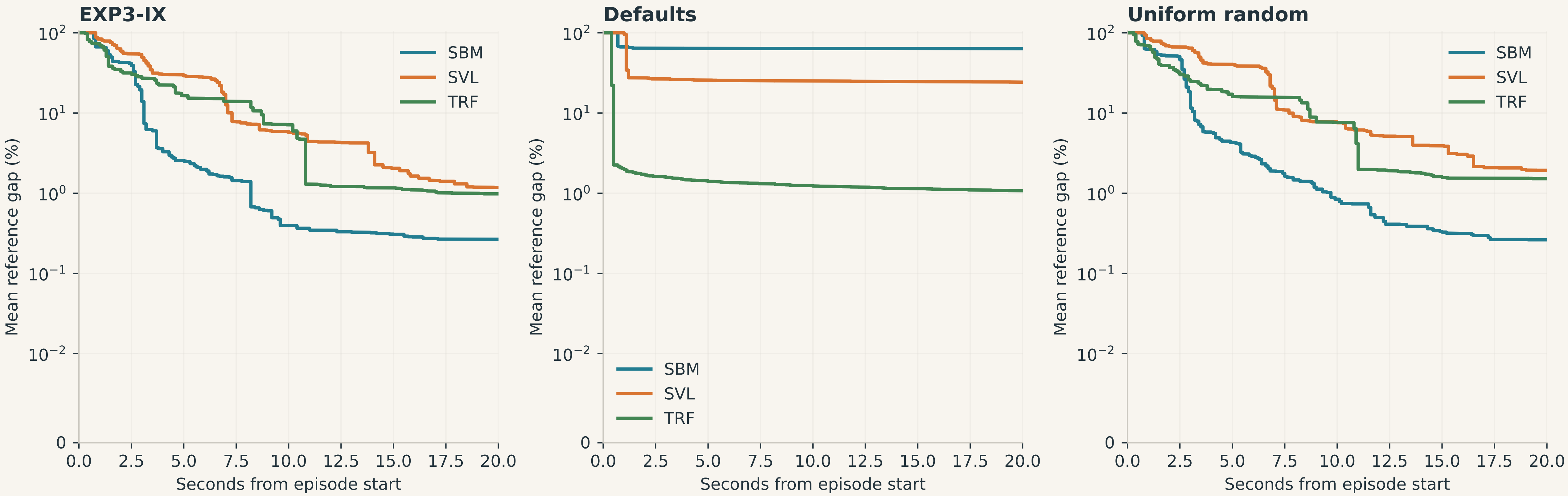
Paired by held-out graph and seed. Positive = EXP3-IX improves quality; negative = the control is better.



These are descriptive paired differences, not confidence intervals or proof of statistical significance.

Quality as the 20-second window unfolds

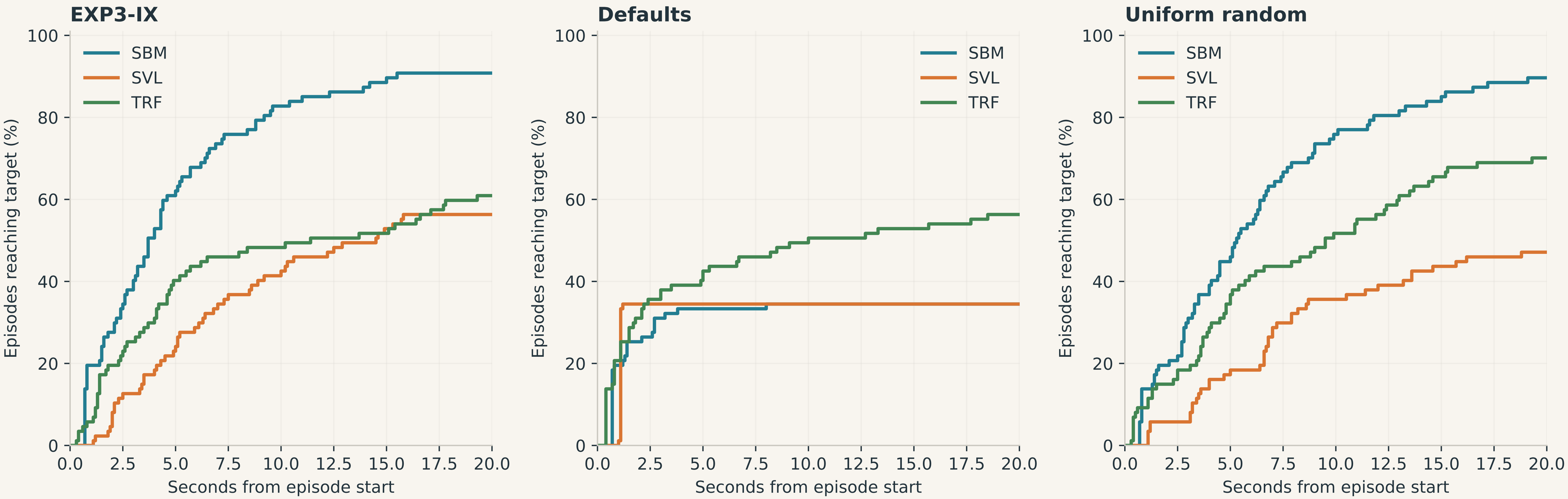
Held-out graphs only. Incumbents reconstructed from saved on-time improvement curves; lower is better.



Gap axis is linear below 0.01%, logarithmic above. Zero-cut fallback is available at time zero.

How soon is a good solution available?

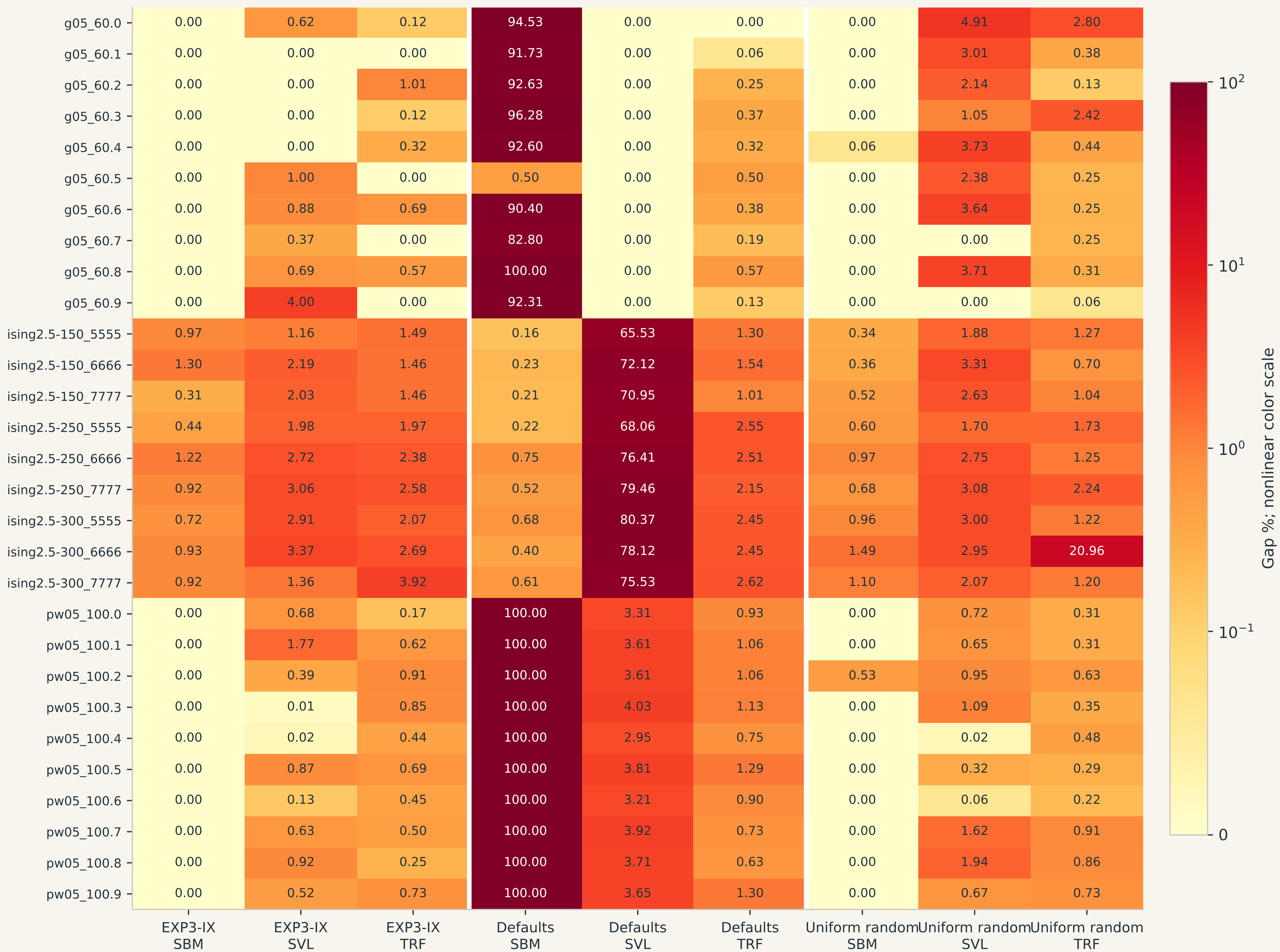
Held-out graphs only. Target: reference gap <= 1%. Unsuccessful episodes remain in the denominator.



Observed first target hit, not statistical restart TTS. Late completions are excluded.

Every held-out problem

Mean reference gap (%) across three seeds. Smaller values are better; each graph has equal weight.



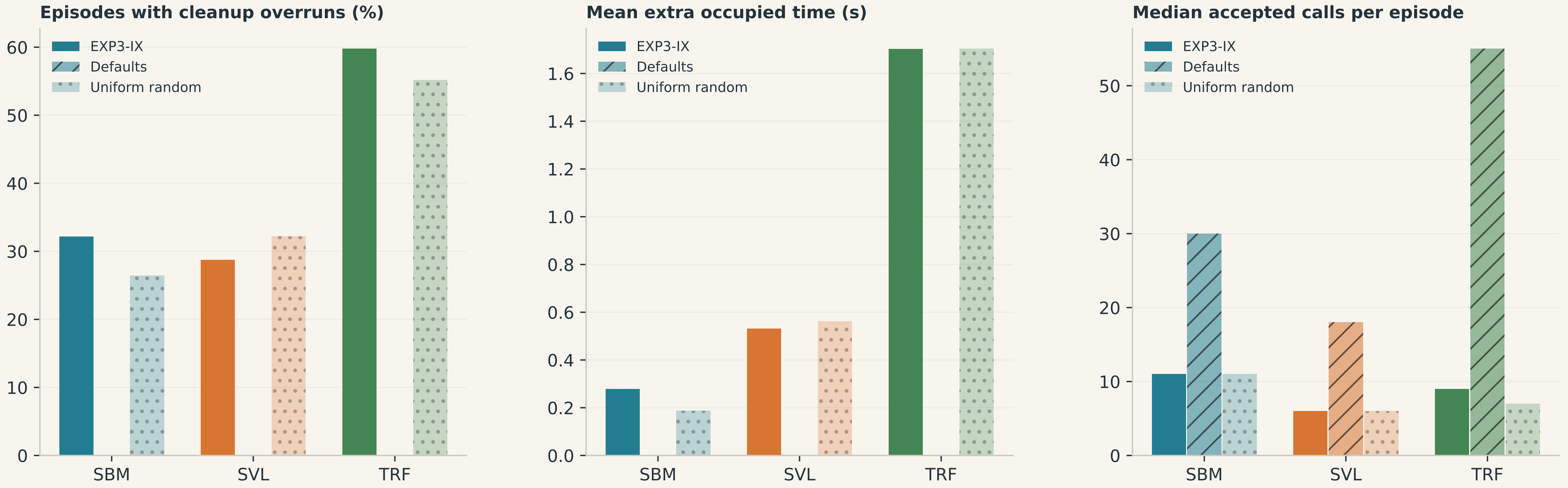
Training history and checkpoint selection

Top: rolling 20-episode training gap, with changing graph mix. Bottom: fixed validation set, one seed.

The training trace is not a fixed-problem learning curve. Only held-out evaluation measures generalization.

Deadline and throughput diagnostics

Held-out episodes. Faster calls can permit more restarts; the quality objective remains best cut by 20 seconds.



An in-flight call may finish after 20 seconds, but its result receives no quality credit. No GPU kernel preemption.